Reducing Stockouts and Overstocking through AI-Driven Inventory Optimization

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Abstract

Effective inventory management is essential for reducing stockouts, minimizing excess inventory, and improving operational efficiency. However, small and medium-sized enterprises (SMEs) often struggle with accurate demand forecasting because they lack large datasets, advanced analytical tools, and technical expertise. This research proposes a lightweight, explainable, and computationally efficient AI-based forecasting framework tailored for SME needs. Using the Walmart Weekly Sales dataset, the study evaluates three demand levels—single store, multi-store, and fully aggregated sales—to replicate SME business scales. A comprehensive modelling pipeline was developed, including baseline statistical methods, SARIMAX, XGBoost, and a hybrid XGBoost–Random Forest model. Results show that XGBoost achieves the highest forecasting accuracy, reaching a MAPE of approximately 1.21% on aggregated data, significantly outperforming traditional models and values reported in prior literature. The study further integrates these forecasts into inventory planning through the calculation of safety stock and reorder points. The proposed approach demonstrates that SMEs can adopt simple, affordable AI systems to make data-driven inventory decisions without requiring large computational resources.

1 Introduction

Inventory management is one of the most important parts of running any business because it directly affects how efficiently a company works, how satisfied customers are, and how much profit it earns. For small and medium-sized enterprises (SMEs), managing inventory properly is not easy since they often have limited money, data, and technical knowledge. Unlike big companies that can afford advanced artificial intelligence (AI) or machine learning (ML) systems, most SMEs still depend on manual work or simple Excel-based forecasting. Because of this, they often face issues like stockouts, overstocking, or inaccurate demand prediction, which leads to loss of sales and wastage of resources.

In today's competitive world, predicting demand accurately has become very important for all types of businesses. Many traditional methods such as the Economic Order Quantity (EOQ) model, Moving Average, and ARIMA (Auto-Regressive Integrated Moving Average) are used for forecasting. However, these methods are not always reliable when the market changes quickly or when there are non-linear patterns in data. Nowadays, AI and ML models like Random Forest, XGBoost, and Neural Networks have shown better performance because they can learn complex relationships from past data and make more accurate predictions.

Even though AI-based inventory systems are very useful, most of the existing ones are developed for large organizations that have a lot of data and strong computing power. Small and medium businesses find it difficult to use them because of high costs, complex systems, and lack of skilled people. So, there is a real need for models that are simple, lightweight, and easy to understand but still give accurate results. These models should also work well with small or incomplete datasets, which is a common situation in SMEs.

This research tries to fill that gap by developing an AI-based inventory optimization model specially designed for small and medium-sized enterprises. The main goal is to create a model that can predict demand more accurately, reduce both stockouts and extra stock, and improve overall efficiency. By using lightweight and explainable machine learning algorithms, this study aims to help SMEs take data-driven decisions and manage their inventory in a smarter and more affordable way.

Main Contributions of this Study

- Development of an **SME-friendly, lightweight XGBoost-based forecasting pipeline** that is accurate, explainable, and feasible on standard laptop hardware.
- Comprehensive **evaluation across three operational scales**—single store, small retail chain, and fully aggregated enterprise—demonstrating robustness under different data availability and demand stability conditions.
- **Integration of demand forecasts into actionable inventory metrics**, including safety stock and reorder points, directly linking predictive accuracy to operational decision-making.
- Empirical **demonstration of state-of-the-art forecasting accuracy** (MAPE as low as 1.21%) using affordable computational resources, outperforming classical models and results reported in prior literature.

2 Literature Review

Inventory management is central to smooth supply-chain operations. Firms must avoid two opposite problems: stockouts, which lead to lost sales and unhappy customers, and overstocking, which increases holding costs and can cause waste. Traditional methods such as EOQ, simple regression and ARIMA work reasonably well in stable situations, but they struggle when demand is volatile, seasonal or affected by sudden events. In recent years, researchers have therefore turned to artificial intelligence (AI) and machine learning (ML) to improve demand forecasting and to support smarter stock decisions.

A comprehensive review by Procir [2022] surveys AI methods for demand forecasting and shows that machine learning models—neural networks, support vector machines and ensemble approaches—often outperform classical statistical techniques, especially when demand patterns are complex or seasonal. The paper summarises empirical comparisons and emphasises that AI methods adapt better to nonlinear behaviour and multiple input features (promotions, holidays, weather), which makes them more suitable for modern retail and e-commerce environments. This work gives strong motivation to replace or augment old forecasting systems with ML tools.

Bhavikatta [2025] focuses specifically on inventory optimisation and presents how various AI techniques can be used to reduce both stockouts and excess inventory. The paper describes methods such as reinforcement learning that can learn ordering policies from past data, and deep models that capture hidden demand patterns. Importantly, Bhavikatta discusses operational metrics (fill rate, holding cost) and shows how AI optimisation can be tuned to business objectives. For this study, this paper is useful because it directly links forecasting improvements to inventory decisions and quantifies the potential benefits.

Kumar et al. [2024] present applied work that combines forecasting and inventory control using a mix of methods—regression models, tree-based ensembles and neural networks—and they demonstrate improved forecasting accuracy and turnover in case studies of retail datasets. Unlike purely theoretical reviews, their paper explains practical steps: feature engineering (lagged sales, promotions), model selection, and how forecasts feed into reorder calculations. They also provide comparative results that show ensemble models often give a good balance of accuracy and interpretability. This paper is especially helpful because it suggests low-complexity pipelines that a small firm could try before moving to heavier solutions.

Radhakrishnan and Shankar [2024] emphasise predictive analytics as a business capability, and they include multiple short case studies where forecasting and demand signals were used to reduce wastage and improve service levels. Their work is more practitioner-oriented: it explains how to integrate forecasts into the planning process, how to measure forecast value, and how to set simple safety-stock rules using forecast error statistics. This practical orientation is important for SMEs, because it shows that benefits come not only from model accuracy but also from sensible operational deployment and continuous model updates.

Alongside technical gains, ethical and sustainability considerations are increasingly important. The study in Discover [2025] examines risks such as data privacy, decision transparency and unintended consequences like overproduction if optimisation is profit-only driven. It argues that AI implementations must include explainability and checks for fairness to avoid harming workers or creating environmental waste. This ethical view complements the technical papers and is necessary when proposing AI solutions for real businesses.

Taken together, these five works show that AI and ML offer clear improvements in forecasting and inventory control, but most solutions target large firms with abundant data and resources. The notable gap is that few studies provide lightweight, low-cost, and easy-to-deploy approaches for small and medium enterprises. This research therefore explores simple, explainable ML pipelines and practical deployment steps that make AI inventory optimisation accessible to smaller businesses.

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3 Methodology

The objective of this research is to develop a lightweight, accurate, and explainable AI-based forecasting model that can support small and medium-sized enterprises (SMEs) in making inventory decisions. The proposed approach predicts weekly product demand and uses these forecasts to compute inventory metrics such as safety stock and reorder points.

The methodology consists of the following major steps: data preparation, feature engineering, model development, model evaluation, and inventory optimisation.

3.1 Why the Walmart Dataset and How It Represents SMEs

This study uses the **Walmart Weekly Sales Dataset** from Kaggle, which contains multi-store sales data along with external economic variables such as holiday indicators, temperature, fuel price, CPI, and unemployment. The dataset was chosen for three reasons:

- It is a **real-world retail dataset** with weekly frequency, which matches the planning cycles used by many SMEs.
- It includes **multiple stores**, allowing the same modelling pipeline to be tested at different granularities: a single outlet, a small network of outlets, and a fully aggregated business.
- It contains **external drivers** (economic and holiday variables), which are useful for building realistic forecasting models with exogenous inputs.

To make the dataset comparable to SME scenarios, the analysis is structured into three levels:

1. **Store 1**: represents a single SME outlet with its own weekly demand.
2. **Stores 1–10**: aggregates the first ten stores, representing a small chain of outlets owned by an SME.
3. **Aggregated (all stores)**: sums sales across all 45 stores to obtain a single high-level time series that mimics a medium-sized business tracking total demand.

By applying the same models to these three levels, the study shows how the approach can scale from one small shop up to a larger SME-like network.

3.2 Data Collection

The Walmart dataset provides:

- Weekly sales figures per store,
- Holiday indicators,
- Fuel price,

- Consumer Price Index (CPI),
- Unemployment rates.

For each of the three levels (Store 1, Stores 1–10, Aggregated), weekly sales and corresponding external variables were constructed from this raw data. For the multi-store and aggregated series, weekly sales are summed across stores, the holiday flag is taken as the maximum (if any store has a holiday, the week is treated as a holiday), and continuous variables are averaged.

3.3 Data Preprocessing

Data Validation: The dataset was checked for missing values, duplicate entries, and incorrect formats. No significant data quality issues were identified; duplicate rows were explicitly removed.

Date Conversion and Sorting: The `Date` column was converted to `datetime` using day-first parsing and sorted to ensure chronological order for time-series analysis.

Construction of Three Time Series: From the cleaned data, the following series were built:

- **Store 1**: weekly totals for Store 1 only.
- **Stores 1–10**: weekly totals for Stores 1–10 combined.
- **Aggregated**: weekly totals across all 45 stores.

Each series has 143 weekly observations, covering 2010–2012.

Feature Engineering: For each of the three series—Store 1, multi-store, and aggregated—the same set of time-dependent and external features was created:

- **Lag features:** past 6 weeks of sales (`lag_1` to `lag_6`),
- **Rolling averages:** 2-week, 3-week, 4-week, and 5-week moving averages,
- **Calendar features:** month and week-of-year (ISO week),
- **External variables:** holiday flag, temperature, fuel price, CPI, unemployment.

Rows with incomplete lag/rolling values at the beginning of each series were dropped so that all training examples are fully specified.

3.4 Model Selection

This study evaluates four main model families:

1. **Baseline statistical models**:
   - **Naive**: forecast equals the previous week's sales.
   - **4-week Moving Average**: forecast is the mean of the last four weeks.
2. **SARIMAX** (Seasonal ARIMA with exogenous variables): a classical time-series model with yearly seasonality and external regressors.
3. **Machine learning model (XGBoost)**: a gradient-boosted tree ensemble regressor.
4. **Hybrid model (XGBoost + Random Forest residual correction)**: a two-stage ML model where Random Forest learns the remaining structure in XGBoost's residuals.

Why XGBoost? XGBoost was chosen because it:

- captures nonlinear relationships and interactions between features,
- handles mixed feature types (lags, rolling means, calendar and economic variables),
- is relatively fast and memory-efficient, which is important for SMEs with limited computing resources,
- offers feature importance scores that improve interpretability.

Why a Hybrid Model? The hybrid XGBoost + Random Forest model is included to test whether a second ML model can further reduce error by learning patterns that XGBoost might miss. Random Forest is robust to noise and can approximate complex functions without heavy hyperparameter tuning. Keeping both models allows us to:

- compare a single strong learner (XGBoost) against an ensemble-of-ensembles (Hybrid),
- check robustness: if XGBoost overfits, the residual model may correct systematic bias,
- demonstrate a generic pattern that SMEs can reuse: "primary forecaster + residual corrector".

3.5 Model Training

For each of the three datasets (Store 1, Stores 1–10, Aggregated), the last 26 weeks were reserved as the test set. All earlier weeks formed the training set. This chronological split prevents data leakage and mimics real-world forecasting, where future weeks must be predicted from past information only.

Baselines: Naive and 4-week Moving Average forecasts were generated directly from the full weekly sales series and aligned with the test indices. Their error metrics were computed on the test weeks after dropping any initial missing values.

SARIMAX: For each dataset, a SARIMAX(1,1,1)(1,1,1)$_{52}$ model was fitted with exogenous regressors (holiday, temperature, fuel price, CPI, unemployment, month, week-of-year). The seasonal period of 52 corresponds to yearly weekly seasonality. Although this structure is standard in the literature, on some of the shorter series (especially Store 1) the model showed numerical instability, which is discussed later.

XGBoost: For each dataset, an XGBoost regressor was tuned using `TimeSeriesSplit` with 4 folds and `GridSearchCV` to minimise mean absolute error. The grid searched over the number of trees, tree depth, learning rate, subsampling, and column sampling. The best estimator for each dataset was then retrained on the full training period and used to forecast the 26 test weeks.

Hybrid Model: Given the best XGBoost model, residuals were computed as the difference between actual and predicted sales on the training data. A Random Forest regressor (300 trees, depth 5) was then fitted to predict these residuals from the same feature set. On the test set, the final hybrid forecast is:

$$\hat{y}_{\text{hybrid}} = \hat{y}_{\text{XGB}} + \hat{\epsilon}_{\text{RF}}.$$

This procedure was repeated for the Store 1, Stores 1–10, and Aggregated datasets.

3.6 Model Evaluation

All models were evaluated using three standard error metrics:

- **Mean Absolute Error (MAE)**: average size of forecast error, in units of weekly sales.
- **Root Mean Squared Error (RMSE)**: square-root of the mean squared error, which penalises larger errors more heavily.
- **Mean Absolute Percentage Error (MAPE)**: mean absolute percentage difference between actual and predicted values, expressed in percent.

These metrics make it possible to compare performance across different models and with results reported in previous studies.

Inventory metrics—standard deviation of forecast errors, safety stock, and reorder point—were derived from the hybrid model for each dataset, as explained in Section 4.4.

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4 Implementation

The forecasting pipeline was implemented in Python using libraries such as Pandas, NumPy, Matplotlib, Seaborn, Statsmodels, and Scikit-Learn. XGBoost and Random Forest were implemented using the `xgboost` and `sklearn` libraries respectively.

4.1 Data Preparation

Raw Walmart sales data were loaded, cleaned, and converted into the three weekly time series (Store 1, Stores 1–10, Aggregated). Exploratory analysis included distribution plots, correlation heatmaps and seasonal decomposition to visually inspect trends and holiday spikes.

For each time series, lagged sales, rolling means, calendar features and external variables were engineered and aligned, resulting in three feature matrices suitable for supervised learning.

4.2 XGBoost Models

For each dataset, the XGBoost regressor used the following main hyperparameters (selected via grid search):

- number of trees between 200 and 400,
- maximum depth between 3 and 5,
- learning rate 0.05 or 0.1,
- subsampling and column sampling ratios of 0.8 or 1.0.

The best models achieved low training error while still generalising well to the 26-week test period. For example, on the aggregated series the test MAPE is approximately 1.21%, substantially better than all baselines and SARIMAX.

4.3 Residual Random Forest and Hybrid Forecasts

The Random Forest residual model was trained on the difference between actual sales and XGBoost predictions. Because these residuals are smaller in magnitude than the original sales, the Random Forest focuses on subtle nonlinear patterns and remaining seasonality that the boosted trees did not fully capture.

On this dataset, the hybrid model delivers performance similar to XGBoost: for the aggregated series it slightly improves MAE and RMSE while keeping MAPE at about 1.21%. This indicates that XGBoost already captures most of the systematic structure, and the residual model acts as a small robustness enhancement rather than a dramatic improvement.

4.4 Inventory Optimisation Outputs

Inventory-related metrics were computed using the hybrid model, which is the most robust and stable forecaster across all three datasets. For each dataset, hybrid forecast errors on the test set were used to compute:

- **Standard deviation of forecast errors** (σ_ϵ): measures how much actual demand typically deviates from the forecast.
- **Average weekly demand** ($\bar{D}$): mean of actual sales over the test period.
- **Safety stock** (SS): a buffer calculated as

  $$SS = z \times \sigma_\epsilon,$$

  where $z = 1.65$ corresponds approximately to a 95% cycle service level.
- **Reorder point** (ROP): the inventory position at which a new order should be placed,

  $$ROP = \bar{D} + SS,$$

  assuming a one-week lead time.

These formulas translate statistical forecast uncertainty into operational quantities that SMEs can use directly when setting reorder thresholds. The results for each dataset are discussed in Section 5.3.

4.5 Scenario Analysis

To explore how external variables affect forecasts, a simple scenario analysis was performed on the aggregated model. Using the last available test week, the unemployment rate was artificially increased by two percentage points, while all other features were kept constant. The XGBoost model then produced a new forecast under this scenario.

The baseline forecast for the last week was approximately 45.64 million, while the scenario forecast increased only to about 45.65 million, a change of around 6,000 units. This tiny difference suggests that, in this dataset, short-term weekly demand is only weakly sensitive to changes in unemployment at the national level. For SMEs, this result indicates that internal factors (recent sales history and seasonality) are more important drivers than macroeconomic indicators in the short term.

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5 Evaluation

This section presents the performance of all forecasting models across the three datasets and compares them with reference results from the literature. It also interprets the inventory metrics in practical terms.

5.1 Model Performance Results

Table 1 summarises the forecasting performance of all evaluated models across the three datasets—Store 1 (single outlet), Stores 1–10 (small chain), and the Aggregated dataset (all stores). Performance is reported using standard error metrics (MAE, RMSE, and MAPE) and is compared with representative ranges from prior literature.

Beyond numerical accuracy, evaluating forecasting models across multiple operational settings is essential for understanding their practical usefulness for small and medium-sized enterprises. Each dataset analysed in this study represents a different decision-making context: a single outlet with limited data (Store 1), a small retail chain with moderate aggregation effects (Stores 1–10), and a medium-sized enterprise operating at an aggregated level (all stores). Assessing model performance across these scenarios allows the analysis to capture how forecasting accuracy, stability, and robustness change as demand becomes smoother and more predictable through aggregation. This multi-level evaluation provides deeper insight into which models are suitable for SMEs at different stages of scale and complexity.

Table 1: Forecasting performance comparison across datasets and literature

Model / Study	MAE	RMSE	MAPE (%)
Store 1 (Single Outlet)			
Naive baseline	91,220.19	116,430.69	5.71
4-week Moving Average	71,200.20	82,644.69	4.53
SARIMAX	Extremely large	Extremely large	Unstable
XGBoost	29,631.06	37,995.15	**1.87**
Hybrid XGB+RF	29,522.22	37,924.55	**1.87**
Stores 1–10 (Small Chain)			
Naive baseline	382,362.78	516,931.63	3.23
4-week Moving Average	321,640.68	387,685.83	2.74
SARIMAX	5,698,640.21	6,525,166.69	49.03
XGBoost	168,824.92	227,769.58	**1.43**
Hybrid XGB+RF	168,825.97	227,774.40	**1.43**
Aggregated (All Stores)			
Naive (last week)	1,462,374.83	2,132,977.00	3.11
4-week Moving Average	1,488,322.72	1,777,827.93	3.19
SARIMAX $(1,1,1)(1,1,1)_{52}$	1,783,549.60	2,191,025.47	3.78
XGBoost (this study)	574,268.18	797,673.84	**1.21**
Hybrid XGB+RF (this study)	**573,203.65**	**795,297.48**	**1.21**
Reference Literature			
ML models (Procir et al., 2022)	–	–	5–15
Tree ensembles (Praveen et al., 2024)	–	–	6–10
Hybrid ML vs. ARIMA (Bhavikatta, 2025)	–	–	10–25% improvement

Several important conclusions can be drawn from Table 1. Across all three datasets, machine learning models—particularly XGBoost and the Hybrid XGBoost+Random Forest model—consistently outperform traditional statistical baselines.

In the Store 1 scenario, XGBoost reduces MAPE from 5.71% (naive) and 4.53% (moving average) to approximately 1.87%, demonstrating its ability to capture nonlinear demand patterns even with limited historical data.

For the multi-store case (Stores 1–10), forecasting becomes more complex due to aggregation across outlets. Although baseline errors decrease slightly because of averaging effects, SARIMAX performs poorly, producing a MAPE of 49.03%. In contrast, XGBoost and the Hybrid model maintain strong performance with a MAPE of 1.43%, highlighting the robustness of tree-based models in multi-store environments.

The aggregated dataset, which best represents a medium-sized enterprise, exhibits the most stable demand structure. Here, XGBoost achieves a very low MAPE of approximately 1.21%, substantially outperforming naive, moving-average, and SARIMAX models. The Hybrid model offers only a marginal improvement in MAE and RMSE, indicating that most systematic demand structure is already captured by XGBoost.

Finally, when compared with published results in the literature, the proposed approach performs significantly better. While prior studies report MAPE values typically between 5–15%, the models in this study consistently achieve errors below 2%. This confirms that a carefully engineered yet lightweight XGBoost-based pipeline can match or exceed the accuracy of more complex forecasting systems while remaining feasible for small and medium-sized enterprises.

Key Insight. XGBoost consistently achieves the lowest forecasting error across all datasets, confirming that a lightweight tree-based model is sufficient for high-accuracy SME demand forecasting.

5.2 Comparison with Literature

According to Procir [2022], machine learning models typically achieve MAPE values between 5–15% on retail demand datasets. Kumar et al. [2024] report MAPE values around 6–10% for tree-based ensemble models in retail forecasting case studies. Bhavikatta [2025] show that hybrid ML models often improve accuracy by 10–25% over classical ARIMA methods.

By contrast, this study's XGBoost and Hybrid models achieve a MAPE of about 1.21% on the aggregated series and 1.4–1.9% on the smaller series (Store 1 and Stores 1–10). Even the simple 4-week moving average baseline is already competitive, but the ML models consistently halve the error. This demonstrates that a lightweight ML approach with well-designed features can deliver state-of-the-art performance while remaining explainable and computationally affordable.

5.3 Inventory Planning Results

Table 2 summarises the key inventory-planning outputs derived from the hybrid forecasting model for all three datasets. These metrics convert statistical forecast errors into operational inventory parameters, allowing uncertainty in demand to be explicitly accounted for in planning decisions. By linking forecast accuracy to safety stock and reorder points, the table provides SMEs with a clear and practical framework for translating predictive performance into day-to-day inventory control.

Table 2: Inventory planning metrics derived from hybrid forecasts

Dataset	Error Std.	Avg. Demand	Safety Stock	Reorder Point
Store 1	38,603.45	1,575,656.94	63,695.70	1,639,352.64
Stores 1–10	224,748.59	11,763,770.48	370,835.17	12,134,605.65
Aggregated (All Stores)	727,729.18	46,874,778.91	1,200,753.15	48,075,532.07

Interpretation.

The error standard deviation reflects forecast uncertainty and increases with business scale, as larger operations naturally exhibit higher absolute demand variability. Safety stock represents the additional inventory required to maintain a 95% service level and grows proportionally with forecast uncertainty.

For example, the aggregated dataset—representing a medium-sized enterprise—requires approximately 1.2 million units of safety stock above average weekly demand to reliably prevent stockouts. The reorder point combines expected demand and safety stock and indicates the inventory level at which replenishment should be triggered, assuming a one-week lead time.

From an SME perspective, these metrics are directly actionable. A single-store operation can use the Store 1 values for weekly ordering decisions, while small chains and medium-sized firms can apply the same framework at aggregated levels to balance service quality and holding costs.

5.4 Scenario Analysis

The scenario analysis with a +2% shock in unemployment on the last test week showed only a small change in forecasted demand (about 6,000 units on a base of 45.6 million).This suggests that, in this dataset:

- macroeconomic variables such as unemployment have limited short-term impact on weekly sales,
- internal factors (recent sales history, seasonality and holidays) dominate the model's decisions,

- SMEs can prioritise getting clean transactional data and basic calendar features before investing effort in collecting complex external indicators.

However, the same scenario-analysis approach can be reused by SMEs whenever they suspect that an external factor (for example, a promotion or a local event) might significantly affect demand. By systematically adjusting input variables and observing the resulting forecast changes, decision-makers can better understand demand sensitivity and assess potential risks before implementing operational changes. This makes scenario analysis a lightweight yet valuable decision-support tool for proactive inventory planning.

5.5 Discussion

The results confirm that:

- Classical statistical models such as SARIMAX struggle on some of the shorter series (Store 1 and Stores 1–10), where they show numerical instability or high error, although they perform reasonably on the full aggregated series.
- Simple baselines (naive and moving average) are surprisingly strong, especially on stable aggregated data, but they cannot capture complex seasonal patterns and external influences as effectively as ML models.
- Machine learning models, particularly XGBoost, provide superior forecasting accuracy across all three datasets, with MAPE between 1.2% and 1.9%.
- The hybrid residual model (XGBoost + Random Forest) offers a small robustness improvement and confirms that there is little systematic structure left in the residuals, which is a good sign from a modelling perspective.

Compared to reference literature, this study achieves higher accuracy with a simpler model pipeline, highlighting its relevance for SMEs that need accurate forecasts but cannot invest in very complex or expensive infrastructure. In addition, the results show that aggregation plays a key role in improving forecast stability, as errors consistently decrease from single-store to aggregated demand levels. This suggests that SMEs operating multiple outlets can benefit substantially from centralized forecasting rather than relying on isolated store-level predictions. Finally, the strong performance of XGBoost across all scenarios demonstrates that carefully engineered tree-based models offer an effective balance between accuracy, interpretability, and computational efficiency, making them particularly well suited for real-world SME inventory planning.

6 Conclusion

This research developed a lightweight and explainable AI-based forecasting framework designed specifically for SMEs. Using real-world Walmart data transformed into three SME-like scenarios (single store, small chain, and fully aggregated business), the study evaluated baseline statistical models, SARIMAX, XGBoost, and a hybrid XGBoost + Random Forest residual model.

The key contributions are:

- A practical, implementable forecasting pipeline, based on open-source Python libraries, that requires only moderate computational resources.
- Demonstration that an ML model (XGBoost) can achieve very low MAPE values (around 1.21% on the aggregated series and below 2% on smaller series), clearly outperforming classical models and reference studies.
- A hybrid residual model that confirms the robustness of XGBoost and provides a generic pattern that SMEs can reuse when they wish to further reduce systematic forecast errors.
- Integration of forecasting outputs into an inventory optimisation framework that computes safety stock and reorder points from forecast error statistics, giving SMEs directly usable guidelines to reduce stockouts and avoid overstocking.
- A simple scenario-analysis procedure that allows decision-makers to test "what-if" changes in external variables, improving their understanding of how sensitive demand is to economic conditions.

Overall, the proposed method enables SMEs to use data-driven strategies to reduce stockouts, avoid excess inventory, and improve operational efficiency without requiring large budgets or specialist teams. Future work may include incorporating promotional and price data, item-level forecasting, and reinforcement learning for automated ordering strategies that directly optimise business objectives such as service level, cost and profit.

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